

Continuum Mechanics
class #14
10/01/2026

Motions

Definition: Let $B \subseteq \mathbb{E}^3$ be given. The smooth mapping

$$\vec{\phi}: B \times [0, \infty) \rightarrow \mathbb{E}^3$$

is called a motion. For each fixed $t \in [0, \infty)$ we assume that

$$\vec{\phi}_t := \vec{\phi}(\cdot, t): B \rightarrow \mathbb{E}^3$$

is an admissible deformation, and generally

$$\vec{\phi}_0 = \vec{\phi}(\cdot, 0): B \rightarrow B$$

is the identity map. We call

$$B_t := \vec{\phi}_t(B) = \vec{\phi}(B, t)$$

The deformed configuration at time $t \in (0, 1)$. ///

Remark: The assumption of continuity implies that, if no holes or ruptures are present in B , then no holes or ruptures are present in $\vec{\phi}(B, t)$, for all $t \in (0, \infty)$. ///

Remark: Since $\vec{\Phi}_t$ is admissible at every t , it has an inverse function

$$\vec{\Psi}(\cdot, t) = \vec{\Psi}_t := \vec{\Phi}_t^{-1} : B_t \rightarrow B$$

for all $t \in [0, \infty)$. If, for fixed $t \in (0, \infty)$,

$$\vec{x}_t = \vec{\Phi}_t(\vec{X}),$$

then

$$\vec{\Psi}_t(\vec{x}_t) = \vec{X}.$$

By the usual properties of inverse, for any $t \geq 0$,

$$\vec{X} = \vec{\Psi}_t(\vec{\Phi}_t(\vec{X})) = \vec{\Psi}(\vec{\Phi}(\vec{X}, t), t), \quad \forall \vec{X} \in B$$

and

$$\vec{x} = \vec{\Phi}_t(\vec{\Psi}_t(\vec{x})) = \vec{\Phi}(\vec{\Psi}(\vec{x}, t), t), \quad \forall \vec{x} \in B_t. \quad ///$$

Definition: A material field is a function of the points $\vec{X} \in B$, the initial configuration.

A spatial field is a function of the points $\vec{x} \in B_t$, the deformed configuration. ///

Definition: Suppose $\Omega : B \times [0, \infty) \rightarrow \mathbb{E}$ is a material field, and suppose $\vec{\Phi} : B \times [0, \infty) \rightarrow \mathbb{E}^3$ is a motion. The function

$$\Omega_s : B_t \times [0, \infty) \rightarrow \mathbb{E}$$

defined by

$$\Omega_s(\vec{x}, t) := \Omega(\vec{\psi}(\vec{x}, t), t)$$

for all $\vec{x} \in B_t$, for all $t \geq 0$, is called the spatial description of the material field Ω .

Similarly, for any spatial field

$$\Gamma : B_t \times [0, \infty) \rightarrow \mathbb{E}$$

The function

$$\Gamma_m : B \times [0, \infty) \rightarrow \mathbb{E}$$

defined by

$$\Gamma_m(\vec{x}, t) := \Gamma(\vec{\phi}(\vec{x}, t), t)$$

is called the material description of the spatial field Γ . //

We will need to be very careful about derivatives in this subject. In particular, we must be clear about whether a derivative is taken with respect to changes in the material $\vec{X} = (X_1, X_2, X_3)^T$ or spatial $\vec{x} = (x_1, x_2, x_3)^T$ coordinates.

Notation: We will use symbols such as

$$\nabla^x \quad \text{and} \quad \nabla^x$$

to denote derivatives with respect to material (X) and spatial (x) coordinates, resp. //

Time Derivatives

Definition: The total time derivative of a field is the rate of change of that field as measured by an observer who is tracking the motion of each particle in the body.

More explicitly, if $\Omega: B \times [0, \infty) \rightarrow E$ is a material field the total time derivative, denoted $\dot{\Omega}$, is

$$\dot{\Omega}(\vec{X}, t) = \frac{\partial}{\partial t} \Omega(\vec{X}, t)$$

On the other hand, if $\Gamma: B_t \times [0, \infty) \rightarrow E$ is a spatial field the total time derivative, still denoted $\dot{\Gamma}$, is

$$(A.16) \quad \dot{\Gamma}(\vec{x}, t) = [\dot{\Gamma}_m]_s. \quad //$$

Remark: For the case of spatial fields we first find the material description. Second, we take the partial derivative with respect to time, and third, we convert the result to a spatial description. For spatial fields, in general,

$$\frac{\partial}{\partial t} \Gamma(\vec{x}, t) \neq \dot{\Gamma}(\vec{x}, t)$$

as we shall see. ///

Definition: let $B \subset \mathbb{E}$ and suppose $\vec{\phi}: B \times [0, \infty) \rightarrow \mathbb{E}^3$ is a motion. Then

$$\vec{v}(\vec{x}, t) := \dot{\vec{\phi}}(\vec{x}, t) = \frac{\partial}{\partial t} \vec{\phi}(\vec{x}, t)$$

is called the velocity, at time t of the material particle labeled by $\vec{x} \in B$.

Similarly,

$$\begin{aligned} \vec{a}(\vec{x}, t) &:= \ddot{\vec{\phi}}(\vec{x}, t) \\ &= \frac{\partial^2}{\partial t^2} \vec{\phi}(\vec{x}, t) \\ &= \dot{\vec{v}}(\vec{x}, t) \\ &= \frac{\partial}{\partial t} \vec{v}(\vec{x}, t) \end{aligned}$$

is called the acceleration at time t of the material particle labeled by $\vec{x} \in B$. ///

Definition: The spatial velocity and acceleration fields are defined as

$$\begin{aligned} (4.17) \quad \vec{v}(\vec{x}, t) &:= \vec{V}_s(\vec{x}, t), \quad \forall \vec{x} \in B_t, \quad \forall t \geq 0, \\ \vec{a}(\vec{x}, t) &:= \vec{A}_s(\vec{x}, t), \quad \forall \vec{x} \in B_t, \quad \forall t \geq 0. \end{aligned}$$

Result (4.7): Suppose $\vec{\Phi} : B \times [0, \infty) \rightarrow \mathbb{E}^3$ is a motion. let

$$\vec{v}(\vec{x}, t) := \vec{V}_s(\vec{x}, t)$$

denote its spatial velocity. Suppose

$$q : B_t \times [0, \infty) \rightarrow \mathbb{E}$$

$$\vec{w} : B_t \times [0, \infty) \rightarrow \mathcal{V}$$

are arbitrary spatial fields. Then

$$\dot{q} = \frac{\partial}{\partial t} q + \nabla^x q \cdot \vec{v}$$

and

$$\dot{\vec{w}} = \frac{\partial}{\partial t} \vec{w} + (\nabla^x \vec{w}) \vec{v}$$

Proof: Suppose $\mathcal{F} = \{\vec{e}_i\}$ is a frame. Then, recall,

$$\nabla^x q = \frac{\partial q}{\partial x_i} \vec{e}_i, \quad \nabla^x \vec{w} = \frac{\partial w_i}{\partial x_j} \vec{e}_i \otimes \vec{e}_j.$$

By definition,

$$\vec{v}(\vec{x}, t) = \vec{V}_s(\vec{x}, t)$$

Recall

$$\vec{V}(\vec{x}, t) = \frac{\partial}{\partial t} \vec{\Phi}(\vec{x}, t)$$

So

$$\vec{v}(x, t) = \vec{V}(\vec{\Phi}(\vec{x}, t), t)$$

Also by definition (Equation (4.16))

$$\dot{q}(\vec{x}, t) = [\dot{q}_m]_s$$

But

$$q_m(\vec{X}, t) = q(\vec{\phi}(\vec{x}, t), t)$$

Thus, by the chain rule,

$$\begin{aligned}\dot{q}_m(\vec{X}, t) &= \frac{\partial}{\partial t} q_m(\vec{X}, t) \\ &= \frac{\partial}{\partial t} q(\vec{\phi}(\vec{x}, t), t) \\ &= \frac{\partial q}{\partial x_i}(\vec{\phi}(\vec{x}, t), t) \frac{\partial}{\partial t} \phi_i(\vec{x}, t) + \frac{\partial}{\partial t} q(\vec{\phi}(\vec{x}, t), t) \\ &= \frac{\partial q}{\partial x_i}(\vec{\phi}(\vec{x}, t), t) v_i(\vec{x}, t) + \frac{\partial}{\partial t} q(\vec{\phi}(\vec{x}, t), t).\end{aligned}$$

Finally,

$$\begin{aligned}\dot{q}(\vec{x}, t) &= [\dot{q}_m]_s(\vec{x}, t) \\ &= [\dot{q}_m(\vec{X}, t)]_{\vec{X} = \vec{\phi}(\vec{x}, t)} \\ &= \frac{\partial q}{\partial x_i}(\vec{x}, t) v_i(\vec{x}, t) + \frac{\partial}{\partial t} q(\vec{x}, t).\end{aligned}$$

The computation for $\vec{w}$ is similar. ///

Notation: Chain rule calculations are notoriously confusing. I use the following notation to help. Here

$$\underline{\frac{\partial}{\partial t}}$$

means the derivative with respect to the variable in the second position.

Likewise, the notation

$$\frac{\partial}{\partial x_i}$$

means the derivative with respect to the i^{th} coordinate of the variable in the first position.

Corollary: It follows that

$$\begin{aligned} a(\vec{x}, t) &= \dot{\vec{v}}(\vec{x}, t) \\ &= \frac{\partial}{\partial t} \vec{v}(\vec{x}, t) + (\nabla^x \vec{v}) \vec{v}. \end{aligned}$$

Proof: An easy exercise. ///

Remark: This sum of two terms

$$\frac{\partial}{\partial t} \vec{v}(\vec{x}, t) + (\nabla^x \vec{v}) \vec{v}$$

will be very important in the description of the Navier-Stokes equations. ///

Rates of Strain and Rotation

Definition: Let $\Phi: B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion with spatial velocity field $\vec{v}: B_t \times [0, \infty) \rightarrow V$. The rate of strain tensor is

$$L(\vec{x}, t) := \text{sym}(\nabla^x \vec{v}) = \frac{1}{2}(\nabla^x \vec{v} + (\nabla^x \vec{v})^T)$$

The rate of rotation tensor (or just spin tensor) is

$$W(\vec{x}, t) := \text{skew}(\nabla^x \vec{v}) = \frac{1}{2} (\nabla^x \vec{v} + (\nabla^x \vec{v})^T)$$

Example (Exercise 30): Suppose $\vec{x} = \vec{\Phi}(\vec{X}, t)$ is defined by

$$x_1 = \cos(\omega t) X_1 + \sin(\omega t) X_2$$

$$x_2 = -\sin(\omega t) X_1 + \cos(\omega t) X_2$$

$$x_3 = (1 + \alpha t) X_3$$

(a) The inverse $\vec{\Psi}$: It is easy to see that

$$\begin{aligned} \vec{x} &= \begin{bmatrix} \cos(\omega t) & \sin(\omega t) & 0 \\ -\sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 + \alpha t \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \\ &= F(t) \vec{X} \end{aligned}$$

Then,

$$F^{-1}(t) = \begin{bmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & \frac{1}{1 + \alpha t} \end{bmatrix}$$

provided $1 + \alpha t \neq 0$, and

$$\vec{X} = F^{-1}(t) \vec{x}$$

Thus

$$\vec{\psi}(\vec{x}, t) = F^{-1}(t) \vec{x}$$

So

$$X_1 = \cos(\omega t) x_1 - \sin(\omega t) x_2,$$

$$X_2 = \sin(\omega t) x_1 + \cos(\omega t) x_2,$$

$$X_3 = \frac{1}{1 + \alpha t} x_3.$$

(b) The components of $\vec{v}$: Recall

$$\begin{aligned} \vec{V}(\vec{x}, t) &= \frac{\partial}{\partial t} [F(t) \vec{x}] \\ &= F'(t) \vec{x}. \end{aligned}$$

By definition,

$$\begin{aligned} \vec{v}(\vec{x}, t) &= F'(t) [F^{-1}(t) \vec{x}] \\ &= [F'(t) F^{-1}(t)] \vec{x}. \end{aligned}$$

(c) The components of L and W : From above

$$\nabla^x \vec{v} = F'(t) F^{-1}(t)$$

Thus

$$\nabla^x \vec{v} = \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & \frac{\alpha}{1+\alpha t} \end{bmatrix}$$

Therefore

$$L = \text{sym}(\nabla^x \vec{v}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{\alpha}{1+\alpha t} \end{bmatrix}$$

and

$$W = \text{skew}(\nabla^x \vec{v}) = \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The rate of strain only depends upon the stretching of the material in the $\vec{e}_3$ direction.

The rotation only depends upon the rotation rate ω . ///

Remark: See exercise 30 for more interpretation details about L and W .

Definition: let $\vec{\phi}: B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion with spatial velocity field $\vec{v}$. The vorticity of the motion is defined by

$$\vec{\omega} := \nabla^x \times \vec{v}.$$

Proposition: let $\vec{\phi}: B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion with spatial velocity field $\vec{v}$. Then the vorticity is the axial vector of the skew symmetric tensor $2W$, where W is the spin tensor.

Proof: Recall

$$W = \text{shew}(\nabla^x \vec{v}).$$

So

$$2W = \nabla^x \vec{v} - (\nabla^x \vec{v})^T.$$

By definition the curl operator satisfies

$$\begin{aligned} (\nabla^x \vec{v}) \times \vec{a} &= (\nabla \vec{v} - (\nabla \vec{v})^T) \vec{a} \\ &= 2W \vec{a} \end{aligned}$$

By Result 1.3 there is a unique vector $\vec{\omega} \in V$ such that

$$(2W) \vec{a} = \vec{\omega} \times \vec{a}, \quad \forall \vec{a} \in V.$$

$\vec{\omega}$ is called the axial vector of $2W$. We already know that

$$\begin{aligned} \vec{\omega} \times \vec{a} &= (\nabla^x \vec{v}) \times \vec{a} \\ &= (2W) \vec{a} \end{aligned}$$

Therefore

$$\vec{\omega} = \vec{\omega}$$

by uniqueness. ///

Rigid Motion

Definition: A motion $\vec{\phi}: B \times [0, \infty) \rightarrow \mathbb{E}^3$ is called a rigid motion iff

$$\vec{\phi}(\vec{x}, t) = \vec{c}(t) + R(t)\vec{x}$$

where, for each $t \geq 0$,

$$R(t) R^T(t) = I, \quad \det(R(t)) = 1.$$

Proposition: Suppose that $\vec{\phi}: B \times [0, \infty) \rightarrow \mathbb{E}^3$ is a rigid motion. Let $\vec{v}$ denote the spatial velocity field. Then

$$\vec{v}(\vec{x}, t) = \vec{\omega}(t) \times (\vec{x} - \vec{c}(t)) + \dot{\vec{c}}(t),$$

where $\vec{\omega}(t)$ (boldface omega) is a time dependent vector called the spatial angular velocity. Equivalently,

$$(4.18) \quad \vec{v}(\vec{x}, t) = \Omega(t)(\vec{x} - \vec{c}(t)) + \dot{\vec{c}}(t),$$

where $\Omega(t) \in \mathcal{V}^2$ is the unique skew-symmetric second-order tensor with axial vector $\vec{\omega}(t)$. Moreover,

$$\nabla^x \vec{v}(\vec{x}, t) = \Omega(t)$$

So that

$$L = 0 \quad W = \Omega(t)$$

and

$$\vec{w} = \nabla_x^x \vec{v} = 2\vec{\omega}(t).$$

Proof: Exercise. ///